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CHAPTER ONE

1. Introduction

Prostate cancer remains one of the most significant public health challenges affecting men worldwide. According to the latest GLOBOCAN 2024 estimates, prostate cancer accounts for 14.1% of all newly diagnosed cancers in men and is now the second leading cause of cancer-related mortality, responsible for over 6.8% of male cancer deaths globally (Sung et al., 2024). Lifetime risk estimates indicate that approximately 1 in 9 men will be diagnosed with prostate cancer, underscoring the substantial disease burden and the critical importance of early detection (Rawla, 2019). In sub-Saharan Africa, including Uganda, late-stage presentation is common, contributing to disproportionately higher mortality due to limited screening, delayed diagnosis, and resource constraints (Musinguzi et al., 2023).

Despite the clinical importance of early detection, traditional diagnostic approaches face limitations. Prostate-Specific Antigen (PSA) testing, the most widely used screening tool, suffers from low specificity (~20%) and an overall diagnostic AUC of approximately 0.72, often leading to false positives, unnecessary biopsies, and missed early-stage disease (Crawford et al., 2023). These diagnostic challenges highlight the need for more accurate, data-driven tools that can improve classification and reduce diagnostic uncertainty.

Advances in artificial intelligence particularly machine learning (ML) have created new opportunities for enhancing disease prediction and diagnosis. ML algorithms can learn complex, nonlinear patterns from large-scale, multidimensional clinical datasets, outperforming traditional statistical approaches in many medical applications (Esteva et al., 2021). Prior work has demonstrated the effectiveness of ML-based prediction models in oncology: for example, random forests have been used to accurately predict breast cancer using SEER data (Yala et al., 2020), while neural network models have shown superior performance in lung cancer risk prediction compared to classical methods (Kumar et al., 2021).

This project aims to develop a machine-learning-based prediction model for prostate cancer using a large retrospective dataset. By integrating demographic factors, medical history, and laboratory biomarkers from URO Care Hospital, the model seeks to improve diagnostic accuracy beyond conventional screening tools. The developed model will contribute to more reliable risk stratification, earlier detection, and ultimately improved clinical outcomes for men at risk of prostate cancer.

1.1 Problem Statement

The accuracy and reliability of prostate cancer diagnosis are critical for improving patient outcomes and reducing unnecessary medical interventions. However, current diagnostic approaches, primarily PSA testing, digital rectal examination (DRE), and TRUS-guided biopsy face significant limitations. PSA levels often lack specificity, leading to false positives, overdiagnosis, and avoidable biopsies, while DRE and ultrasound findings are highly operator-dependent and may miss clinically significant cancers. These gaps result in delayed detection for some patients and overtreatment for others.

Traditional diagnostic pathways struggle to integrate multiple clinical features and subtle patterns that influence prostate cancer risk. Relying on single biomarkers or subjective assessments limits diagnostic precision. Therefore, there is a growing need for advanced, data-driven approaches capable of capturing complex relationships within clinical variables.

To address these challenges, applying supervised machine learning algorithms offers a promising solution. These models can analyze high-dimensional clinical data to improve predictive accuracy and support early detection. This project therefore aims to develop four supervised machine learning models to classify prostate cancer risk outcomes, evaluate their performance and optimize the most effective model.

1.2 Significance of the study

Early and accurate prediction of prostate cancer is essential for improving patient outcomes, reducing unnecessary procedures, and supporting timely treatment decisions. Traditional diagnostic approaches such as PSA testing and manual interpretation of TRUS and biopsy findings are often inconsistent, time-consuming, and prone to diagnostic uncertainty. The goal of this project is to investigate the use of machine learning approaches to enhance diagnostic accuracy by identifying key clinical predictors and supporting faster, data-driven decision-making. By improving risk stratification and early detection, the study contributes to better clinical management, reduced healthcare burdens, and improved survival outcomes for men at risk of prostate cancer.

1.3 Objectives

1. To analyze the key risk factors associated with prostate cancer among men aged 40 years and above seeking screening services at URO Care Hospital.
2. To develop four supervised machine learning models capable of accurately predicting the likelihood of prostate cancer.
3. To evaluate the performance of the four supervised machine learning predictive models and identify the most effective one.
4. To optimize the best performing supervised machine learning model.

1.4 Rationale

Early detection of prostate cancer is essential for reducing mortality and improving patient outcomes, yet traditional diagnostic methods remain slow, subjective, and often inaccurate. Using supervised machine learning models provides a faster, more consistent, and data-driven way to assess risk, supporting timely clinical decisions. These models can efficiently analyze key clinical variables, reduce diagnostic uncertainty, and enhance screening accuracy, ultimately strengthening prostate cancer management and improving patient care.

1.5 Background of the problem

Prostate cancer remains a major public health concern in Uganda, ranking as the second most common cancer among men and one of the highest recorded across Africa. In recent years, the country has experienced an increasing number of prostate cancer cases, with a large proportion of patients presenting at advanced stages. Statistics from the Uganda Cancer Institute (UCI) reveal that many men do not undergo early screening and often seek medical attention only after symptoms have significantly progressed, contributing to poor treatment outcomes and high mortality rates. Although the UCI recommends a combination of prostate-specific antigen (PSA) testing and Digital Rectal Examination to improve diagnostic accuracy, challenges such as misdiagnosis, late detection, and limited access to health services persist.

The growing burden of prostate cancer highlights the need for stronger, data-driven screening approaches that can accurately identify at-risk individuals and facilitate timely diagnosis. Advances in data science and machine learning offer a unique opportunity to address this gap. In Uganda, machine learning has already been applied in diverse areas, including healthcare, environmental monitoring, and humanitarian response, demonstrating its potential to transform data into actionable insights. Leveraging these analytical capabilities to predict prostate cancer risk could significantly enhance early detection, guide preventive interventions, and reduce the country's prostate cancer mortality rates.

CHAPTER TWO

2. Literature review

2.1 Introduction

This literature review explores the significance of early prostate cancer detection and recent studies on the application of machine learning algorithms in prostate cancer diagnosis and prognosis. It highlights their effectiveness in enhancing predictive accuracy and supporting clinical decision-making, while also identifying key limitations and gaps that require further investigation.

2.2 Risk Factors of Prostate Cancer

The risk factors of prostate cancer can be grouped into either modifiable (those that can be changed or avoided) or non-modifiable (those cannot be changed or practically difficult to change or avoid). The risk factors show significant heterogeneity in their association with prostate cancer development. The modifiable factors include age, genetic changes, family history, testosterone and other sex hormones, and insulin like growth factors (Jatho *et al.*, 2017). It was found that the risk of prostate cancer increases with age, especially after age of 50. More than 80% of prostate cancers are diagnosed in men who are 65 years or older.

Several inherited gene changes can raise prostate cancer risk. Inherited mutations of the Breast cancer genes 1 and 2 (BRCA1 or BRCA2 genes) increase prostate cancer risk in some men. Men with Lynch syndrome, a condition caused by inherited gene changes, have an increased risk for a number of cancers, including prostate cancer. According to Patel and Klein (2009), familial and genetic factors are among the strongest. Some cases of prostate cancer seem to run in some families, which suggests that in some cases there may be an inherited or genetic factor or sharing a particular risk factor. Sex hormones, including testosterone, clearly contribute to the growth and progression of prostate cancer, although there is still debate around the importance of circulating levels of specific hormones or genetic markers in these pathways in prostate cancer.

Non-modifiable factors on the other hand include, diet, overweight and obesity, and physical inactivity. According to Jatho *et al.* (2017), men who eat a lot of red meat (beef, goat, pork etc) or high-fat dairy products appear to have a slightly higher chance of getting prostate cancer. Men who are obese have a higher risk of getting more aggressive prostate cancer. Lastly, consistently using fewer calories than are consumed leads to obesity, which is linked to increased risks of cancers.

Other risk factors include a positive family history, abnormal digital rectal examination (DRE), raised PSA level and ethnicity. For the purposes of this study, risk factors assessed during screening will be analysed to determine their association with prostate cancer. These will include, age, trans rectal ultrasound findings, and PSA levels.

2.3 Prostate Cancer Screening Tests

Screening for prostate cancer relies on multiple approaches, each with distinct strengths and limitations. Symptom assessment, which involves collecting patient history on urinary symptoms, pain, and risk factors, is the first step in identifying individuals at risk. However, early-stage prostate cancer is often asymptomatic, limiting the effectiveness of this approach.

The PSA test measures prostate-specific antigen levels in the blood and is widely used to guide biopsy decisions (Stephan, Ralla & Jung, 2014). While it has improved early detection, its low specificity and negative predictive value mean that elevated PSA levels may result from benign conditions, introducing uncertainty. Statistical and machine learning models, such as logistic regression and artificial neural networks, have been proposed to enhance PSA-based predictions (Hu et al., 2013).

DRE involves palpation of the prostate to detect abnormalities. It is limited by subjectivity, inter-observer variability, and low sensitivity at low PSA levels. TRUS-guided biopsy, the primary imaging method for prostate assessment, allows tissue evaluation but has limited staging accuracy and may under-detect tumors despite technological advances. Prostate biopsy confirms malignancy and provides Gleason grading to guide clinical management, but it is invasive and subject to sampling error due to tumor heterogeneity.

Because no single test is fully reliable, combining PSA, DRE, TRUS, and biopsy results is increasingly recommended to improve detection accuracy. Studies show that integrated approaches increase sensitivity, but optimal combinations are still under investigation, and resource constraints can limit their application in some settings.

2.4 Machine learning for Prostate cancer risk prediction

Recent advancements in machine learning have significantly improved the early detection and clinical management of prostate cancer. Modern studies increasingly use high-dimensional clinical data, MRI imaging, genomic markers, and fused biomarker sets to build accurate predictive models. For example, deep learning models such as convolutional neural networks (CNNs) have demonstrated strong performance in analyzing multiparametric MRI, with accuracies exceeding 90% in lesion detection and Gleason score prediction (Litjens et al., 2024; Aldoj et al., 2023). In clinical datasets, machine learning algorithms such as Random Forests, Gradient Boosting, and Support Vector Machines continue to outperform traditional PSA-based diagnostic models. Recent work by Barua et al. (2023) achieved over 95% accuracy using an ensemble model combining clinical variables and MRI radiomics, while Kazmierski et al. (2022) reported that SVM and XGBoost classifiers significantly reduced unnecessary biopsies by improving risk stratification.

Genomic-based ML approaches are also transforming prostate cancer diagnostics. Studies using gene-expression signatures and machine learning classifiers have shown diagnostic accuracies

above 97%, offering potential for precision oncology (Chen et al., 2023; Wang et al., 2024). Ensemble learning remains one of the most effective strategies, a 2024 review shows that hybrid models integrating multiple algorithms consistently outperform single-model approaches, achieving high sensitivity and specificity in both imaging and non-imaging datasets (Singh & Patel, 2024). These advancements demonstrate that machine learning provides powerful tools for improving diagnostic accuracy, reducing false positives, and supporting personalized decision-making in prostate cancer care.

2.5 Summary

This literature review explores prostate cancer risk factors, screening methods, and recent machine learning advancements. It emphasizes the importance of accurate risk assessment for early detection and improved patient outcomes. Key contributors include age, genetic mutations, family history, hormones, lifestyle, and clinical markers such as PSA, DRE, and TRUS. Machine learning approaches, including Random Forests, SVMs, Gradient Boosting and ensemble models, have further enhanced diagnostic accuracy by integrating clinical, imaging, and genomic data, reducing unnecessary biopsies, and supporting personalized decision-making. However, limitations remain, including low specificity of traditional tests, subjectivity in DRE, under-detection in TRUS, invasive biopsy procedures, and challenges in generalizing machine learning models across diverse populations.

CHAPTER THREE

3. Methodology

3.1 Introduction

This study was conducted at URO Care Hospital, a specialized urology and nephrology healthcare facility in Nansana, Wakiso District, Uganda. The study population comprised men aged 40 years and above who underwent prostate cancer screening between 1st May 2017 and 31st May 2022. The dataset consisted of 1030 participants, including demographic information, clinical findings, and laboratory test results. Patient identifiers were de-identified to ensure privacy and confidentiality throughout the study. The study aimed to identify prostate cancer risk factors, evaluate screening test effectiveness.

3.2 Dependencies

Data analysis and model development were performed using Python, with essential libraries including pandas for data manipulation, numpy for numerical computations, and scikit-learn for machine learning algorithms. Seaborn and matplotlib were used for visualization, while joblib enabled model persistence. These tools ensured a comprehensive approach to data cleaning, analysis, and predictive modeling.

3.3 Dataset Description

The dataset included demographic information such as age, district of residence, family history, lower urinary tract symptoms, DRE findings, PSA levels, TRUS results, and biopsy diagnoses. The outcome variable was the occurrence of prostate cancer, with the remaining variables serving as predictors. Data was analyzed to compare patients with and without prostate cancer and used as input for machine learning models to predict diagnosis.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1030 entries, 0 to 1029
Data columns (total 12 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Age                                       1030 non-null   int64
1   Family History of PC                    1030 non-null   object
2   Frequent Urination                      1030 non-null   object
3   Weak Urine Flow                         1030 non-null   object
4   Frequent Urination at Night/Nocturia    1030 non-null   object
5   Bloody Urine/Heamaturia                 1030 non-null   object
6   Erectile Dysfunction                    1029 non-null   object
7   DRE      B/S/0                          1014 non-null   object
8   Prostae volume on abdominal scan        752 non-null    object
9   PSA before biopsy                       1025 non-null   object
10  Biopsy                                  417 non-null    object
11  Prostate Cancer                         1030 non-null   int64
dtypes: int64(2), object(10)
memory usage: 96.7+ KB

```

Figure 1: Dataset features and datatypes

3.4 Data Exploration and Preprocessing

3.4.1 Data Pre-processing

During the data analysis phase, the patient records dataset was uploaded into Google Colab and the prostate cancer outcome variable was mapped to a binary format, with diagnosed cases labeled as 1 and non-disease cases labeled as 0. Missing values were handled systematically: features with more than 75% missing data were removed, numerical missing values were imputed using the mean, and categorical missing values were imputed using the mode. Binary response fields were standardized by mapping missing responses to 0, “Yes” to Y, and “No” to N, ensuring consistency across the dataset. All columns were cleaned by stripping empty entries and whitespace. Categorical variables were then transformed through one-hot encoding to enable their use in machine-learning algorithms. After preprocessing, the dataset was divided into training and testing subsets, and a StandardScaler from Scikit-Learn was applied to both sets to normalize numerical variables for improved model performance.

```

Columns with missing values:
Erectile Dysfunction          1
DRE      B/S/0                16
Prostae volume on abdominal scan 298
PSA before biopsy             22
Biopsy                        613
dtype: int64
Percentage of missing values:
Erectile Dysfunction          0.0971
DRE      B/S/0                1.5534
Prostae volume on abdominal scan 28.9320
PSA before biopsy             2.1359
Biopsy                        59.5146
dtype: float64

```

Figure 2: columns with missing values and their percentages

```

.. <class 'pandas.core.frame.DataFrame'>
   RangeIndex: 1030 entries, 0 to 1029
   Data columns (total 12 columns):
    #   Column                                     Non-Null Count  Dtype
   ---  ---
    0   Age                                           1030 non-null   float64
    1   Family History of PC                         1030 non-null   object
    2   Frequent Urination                          1030 non-null   object
    3   Weak Urine Flow                             1030 non-null   object
    4   Frequent Urination at Night/Nocturia        1030 non-null   object
    5   Bloody Urine/Heamaturia                     1030 non-null   object
    6   Erectile Dysfunction                         1030 non-null   object
    7   DRE      B/S/0                              1030 non-null   object
    8   Prostae volume on abdominal scan             1030 non-null   float64
    9   PSA before biopsy                           1030 non-null   float64
    10  Biopsy                                       1030 non-null   object
    11  Prostate Cancer                            1030 non-null   int64
   dtypes: float64(3), int64(1), object(8)
   memory usage: 96.7+ KB

```

Figure 3: Dataset without missing values

3.4.2 Feature Selection

Following data cleaning and preprocessing, a structured feature-selection procedure was applied to ensure that only the most informative predictors were included in the final prostate cancer prediction model. The aim was to reduce noise, avoid multicollinearity, and improve model interpretability and performance.

3.4.3 Preliminary Feature Filtering Using Rank Correlation

A rank-based correlation analysis (Spearman's ρ) was first performed to evaluate the monotonic relationship between top ten input feature and the binary prostate cancer outcome. Features with a correlation coefficient below 0.30, such as *district of residence*, were discarded. This threshold ensured that only variables with meaningful and independent associations with the target variable advanced to the next analysis stage.

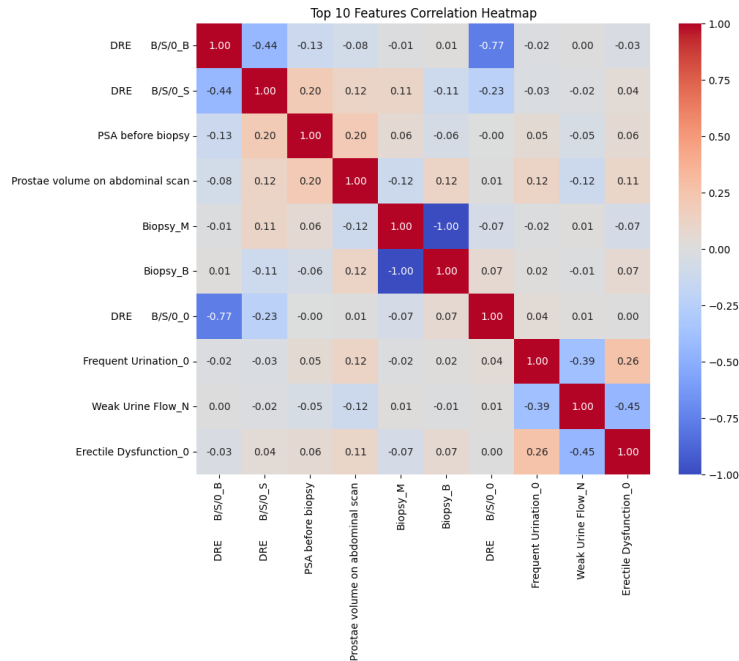


Figure 4: Rank correlation for the top 10 features

3.4.4 Exploratory Feature Distribution Using Pair Plot Visualization

The remaining features were visualized using the Seaborn statistical graphics library (Waskom, 2021). Pairplots were used to assess feature distributions, detect outliers, and observe preliminary relationships between PSA before Biopsy, Prostate Volume on abdominal scan and prostate cancer status.

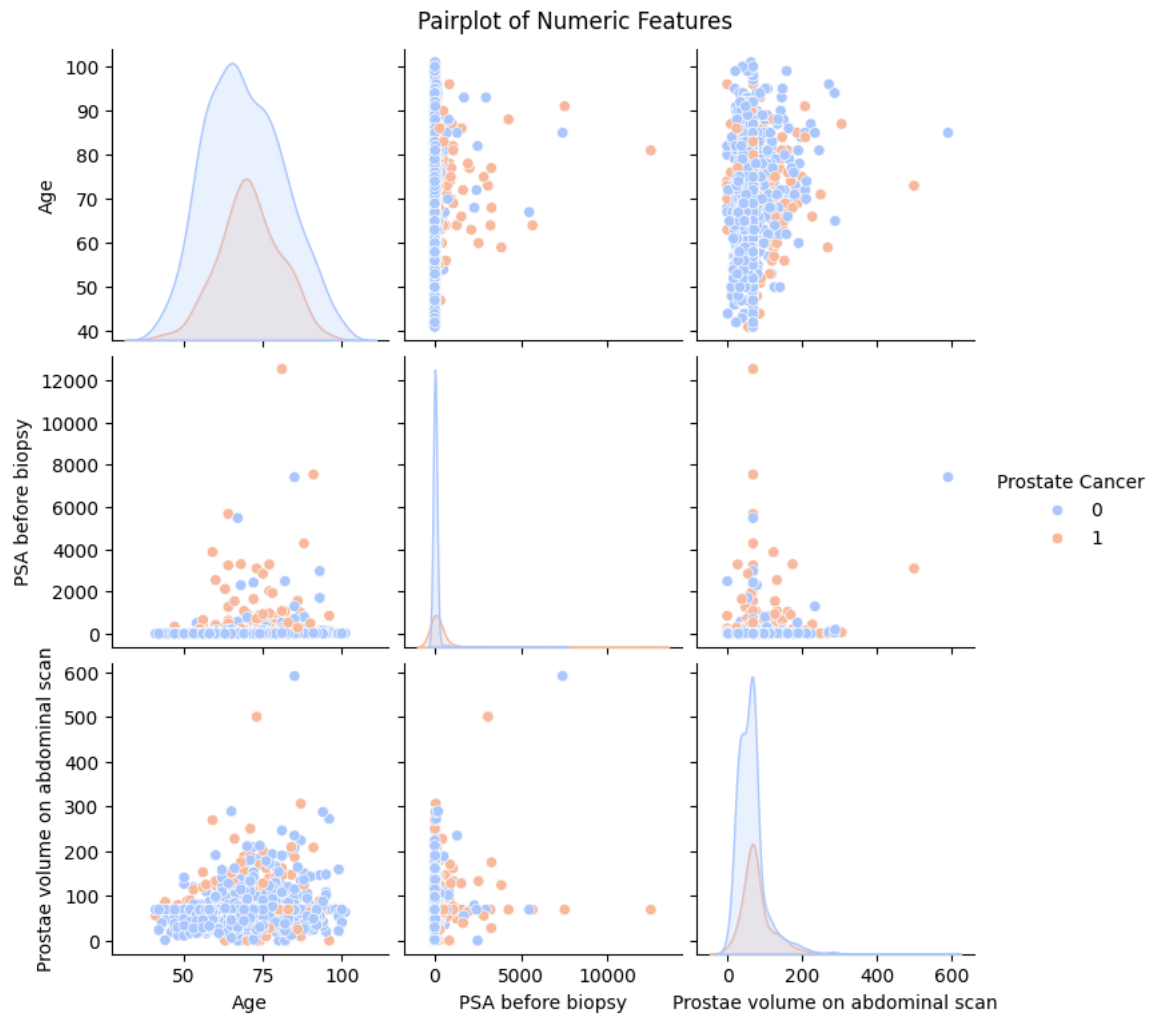


Figure 5: Heat map for numerical features and target variable

4. Implementation and Testing

4.1 Introduction

This chapter details the steps taken to implement, train, and test various Super-vised Machine learning models, including logistic regression, Decision trees, K-nearest neighbor and Random Forest. It also explains how different performance evaluation metrics like recall, accuracy, ROC-AUC, precision, and F1 score were utilized to assess the models' performance. Furthermore, I detail the different techniques used in optimizing model performance.

4.2 Model Development

4.2.1 Logistic Regression

During the data preprocessing phase, Categorical variables were converted into a binary matrix using one-hot encoding. Numerical variables were standardized and to ensure they contribute equally to the model and improve the con-vergence speed.

The initialization of key parameters followed this. These parameters include the penalty, the inverse of regularization strength, the solver, and the maximum number of iterations. The logistic regression model was trained using the training set, where it learned the weights for each feature by minimizing the loss function, commonly binary cross-entropy for binary classification tasks. For predictions, the logistic regression model calculated the probability of each instance in the test set belonging to a particular class using the logistic function. The instance was then assigned to the class with the highest probability.

4.2.2 K- Nearest Neighbors

The K-Nearest Neighbors algorithm was implemented using the scikit-learn library in Python. Data preprocessing included scaling numerical variables to ensure that all features contributed equally to distance calculations, and encoding categorical variables as needed. A KNN classifier was initialized with parameters such as the number of neighbors (k), distance metric, and weighting method. The model was trained by storing the training instances and using distance-based voting among the k nearest neighbors to classify new samples. Model performance was evaluated on the testing set using metrics including accuracy, precision, recall, and F1 score. This process enabled an effective and interpretable implementation of KNN for classification tasks.

4.2.3 Decision trees

The Decision Tree algorithm was implemented using the scikit-learn library in Python. The data preprocessing involved one-hot encoding for categorical variables and scaling for numerical variables where necessary. A Decision Tree classifier was initialized with parameters such as maximum depth, minimum samples per leaf, and splitting criterion. The model was trained on the

training set by recursively partitioning the data based on feature values to create a tree structure that separates classes. The model's performance was evaluated on the testing set using metrics such as accuracy, precision, recall, and F1 score. This approach provided a comprehensive implementation of the Decision Tree algorithm for classification tasks.

4.3.4 Random Forest

The Random Forest algorithm was implemented using the scikit-learn library in Python. The data preprocessing involved one-hot encoding for categorical variables, scaling for numerical variables. A Random Forest classifier was then initialized with specific parameters such as n estimators, max depth, and criterion. The model was trained on the training set by creating multiple decision trees using bootstrap samples, with aggregated predictions to enhance accuracy and control over fitting. The model's performance was evaluated on the testing set using metrics like accuracy, precision, recall, and the F1 score, with cross-validation ensuring robustness. Additionally, the feature importance attribute was utilized to identify the most significant features, aiding in model interpretation. This thorough process ensured an effective implementation of the Random Forest algorithm for classification tasks.

4.3.5 Voting Ensemble Classifier

To enhance predictive accuracy and leverage the strengths of multiple algorithms, a Voting Ensemble classifier was developed using the scikit-learn library in Python. The ensemble combined base models, including Random Forest, Decision Tree, and K-Nearest Neighbors. Preprocessing steps ensured that numerical features were scaled and categorical variables were properly encoded. The ensemble was configured for soft voting, which averages predicted class probabilities. The model was trained on the training set and evaluated on the testing set using accuracy, precision, recall, and F1 score. This method allowed the model to make more reliable and stable predictions by aggregating insights from multiple classifiers.

4.4 Model Optimization

To select the best model, hyperparameter tuning was performed to identify the optimal set of hyperparameters for each developed learning algorithm. This was achieved using the Grid Search method, a widely used approach that exhaustively searches through a manually specified subset of the hyperparameter space. For each combination of hyperparameter values, a model was trained and evaluated using k-fold cross-validation, which splits the training data into k subsets (folds) to ensure robust performance assessment and reduce the risk of overfitting. The process was repeated for all combinations of hyperparameters, and the combination that produced the best cross-validated performance metrics was selected as the optimal set for the model.

4.5 Result Evaluation

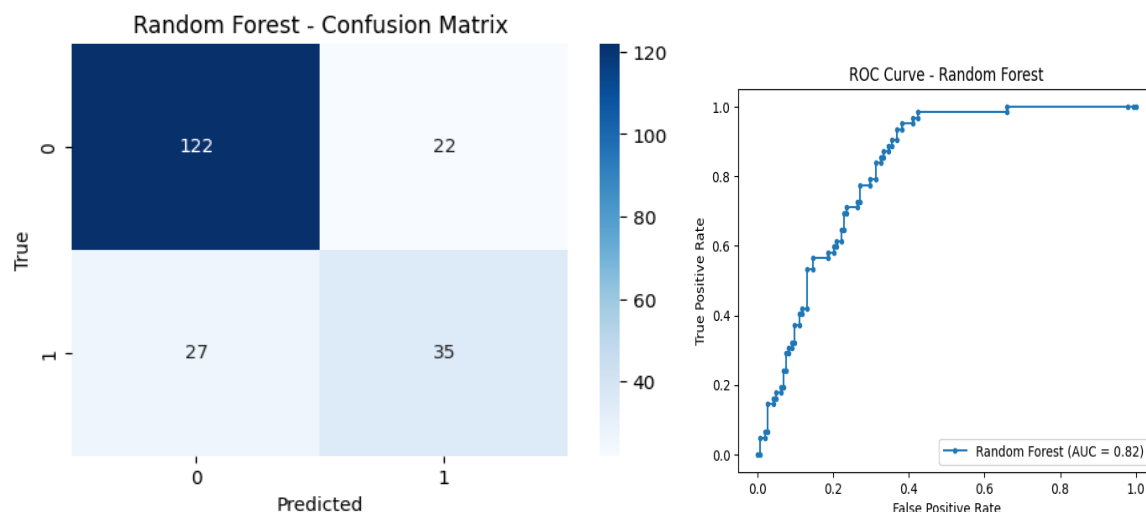
In the evaluation phase, the performance of the four machine learning algorithms; logistic Regression, Decision trees, Random Forest, and K-nearest Neighbors was assessed across various metrics including F1 score, ROC-AUC, Recall, Accuracy, and Precision.

These metrics provided a comprehensive understanding of each model's classification accuracy, sensitivity to positive cases, discrimination ability, and overall correctness of predictions. The evaluation process facilitated the selection of the most suitable model for the classification task based on its performance across these key metrics.

4.6 Results and Discussions

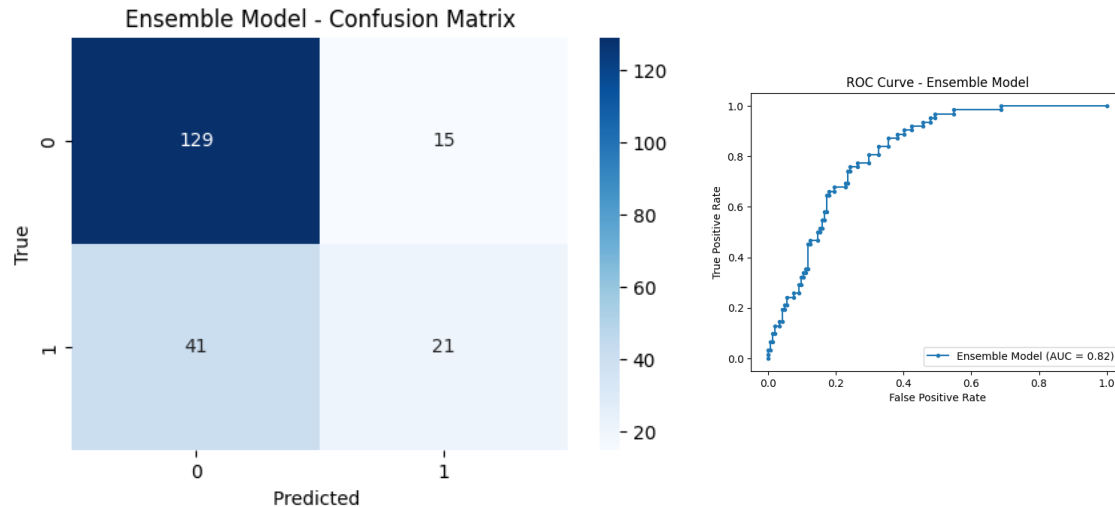
4.6.1 Random Forest

The Random Forest model delivered the strongest performance among all evaluated algorithms, with an accuracy of 0.76, precision of 0.82, recall of 0.85, F1-score of 0.83, and an AUC of 0.82. These results indicate that the model balances sensitivity and specificity well, making it highly suitable for distinguishing between patients with and without prostate cancer. Similar trends have been observed in recent clinical studies, where Random Forest consistently outperforms linear models due to its ability to capture complex interactions among predictors such as PSA levels, DRE findings, and TRUS characteristics. The high AUC demonstrates strong discrimination capacity, reinforcing Random Forest as a reliable tool for clinical decision support in prostate cancer diagnosis.



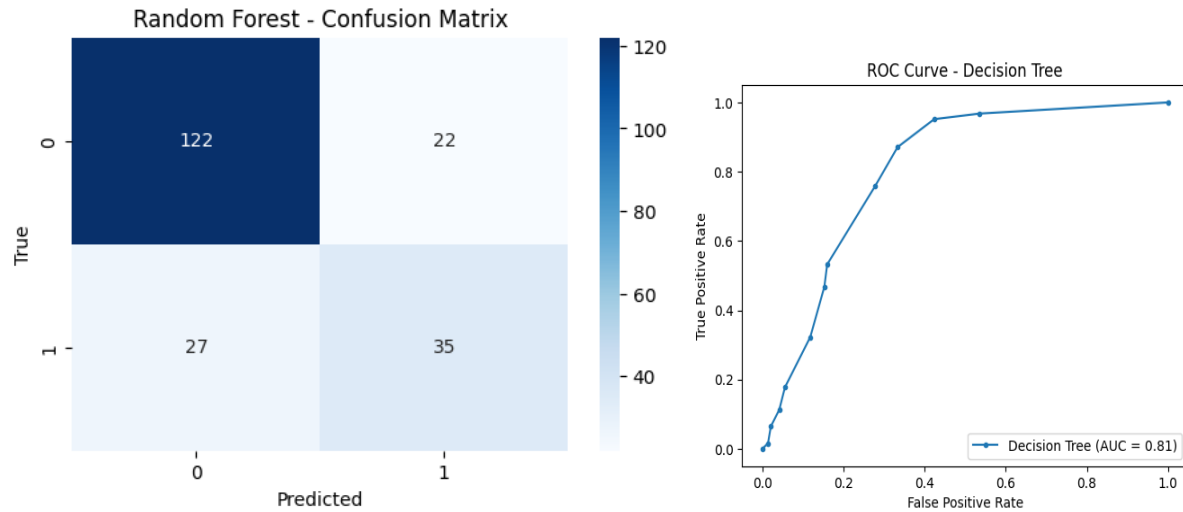
4.6.2 Voting Ensemble

The Voting Ensemble, which combined predictions from multiple base models, produced an accuracy of 0.73, precision of 0.76, recall of 0.90, F1-score of 0.82, and an AUC of 0.82. Despite having slightly lower accuracy than Random Forest, its AUC value indicates strong discriminative performance. This suggests that blending models helps capture complementary strengths, though not enough to surpass the robustness of Random Forest. The ensemble's stable recall highlights its usefulness for screening scenarios where detecting cancer cases is prioritized.



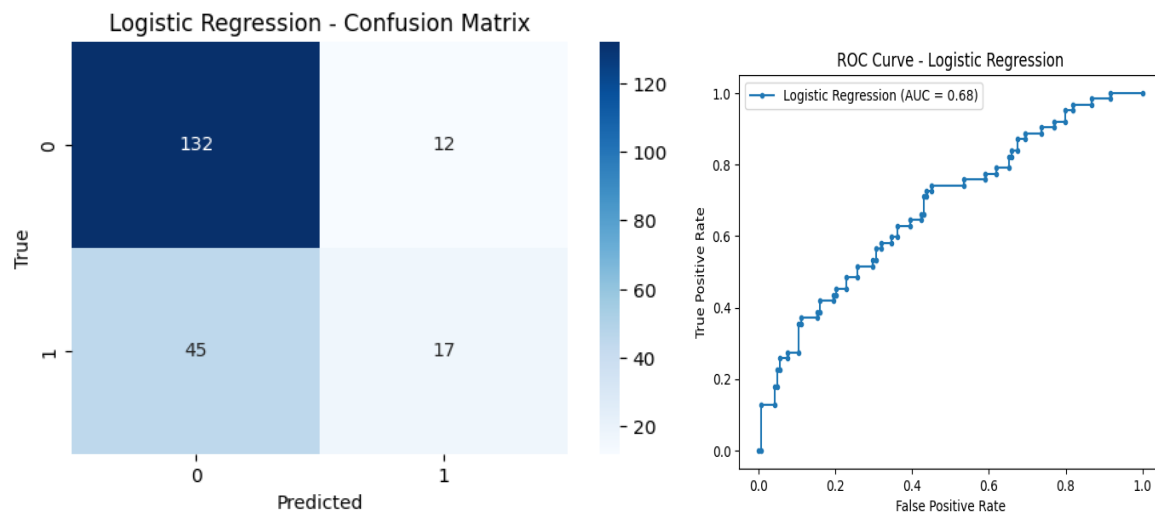
4.6.3 Decision Tree

The Decision Tree model achieved an accuracy of 0.73, precision of 0.79, recall of 0.85, F1-score of 0.82, and an AUC of 0.81. Although its performance was slightly lower than that of Random Forest, it still offered solid predictive capability. Its interpretability makes it valuable for understanding key predictors influencing prostate cancer risk, such as age and PSA levels. However, the lower AUC compared to Random Forest suggests that relying on a single tree limits its ability to generalize well to complex clinical patterns.



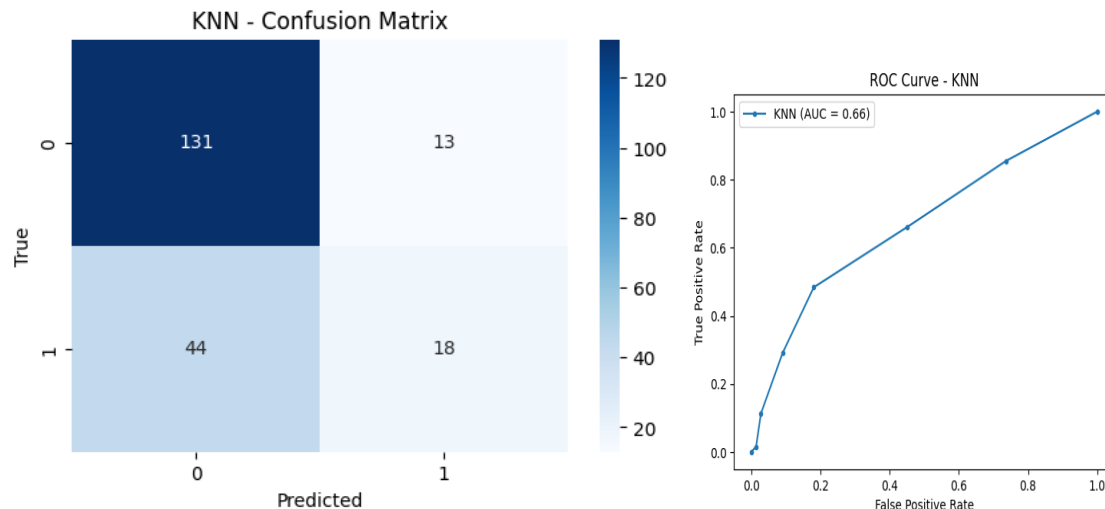
4.6.4 Logistic Regression

Logistic Regression achieved an accuracy of 0.72, precision of 0.75, recall of 0.92, F1-score of 0.82, and an AUC of 0.68. As expected for a linear classifier, its performance was more modest due to the nonlinear relationships inherent in prostate cancer predictors. Nonetheless, its simplicity and interpretability make it a practical baseline model for understanding how risk factors such as PSA and age contribute to cancer probability. The relatively low AUC suggests limitations in distinguishing between cancer and non-cancer cases.



4.6.5 K-Nearest Neighbors (KNN)

The KNN model recorded an accuracy of 0.72, precision of 0.75, recall of 0.92, F1-score of 0.82, and an AUC of 0.66. While its accuracy was comparable to Logistic Regression, its lower AUC indicates weaker discriminative power. KNN's performance is influenced by data scale and density, and the variability within clinical measurements such as PSA and TRUS findings may reduce its effectiveness. These results suggest that KNN is less suited for prostate cancer prediction compared to tree-based models.



4.6.6 Feature Importance

Finally, to assess the predictive power of individual features in classifying prostate cancer risk, I utilized the built-in Feature Importance mechanism of the trained Random Forest model. This method, often referred to as Gini Importance or Mean Decrease in Impurity (MDI), quantifies a feature's contribution to the model's overall performance. PSA levels is ranked as the dominant predictor, PSA accounted for over 42% of the combined importance. This confirms that the quantitative measurement of the Prostate-Specific Antigen is the most critical factor for the model in distinguishing between risk classes. Age is the second most important feature, reflecting the epidemiological fact that age is a fundamental, non-modifiable risk factor for prostate cancer. and prostate volume. The model integrates these primary indicators with findings from the DRE and previous biopsy results to build a comprehensive risk classification.

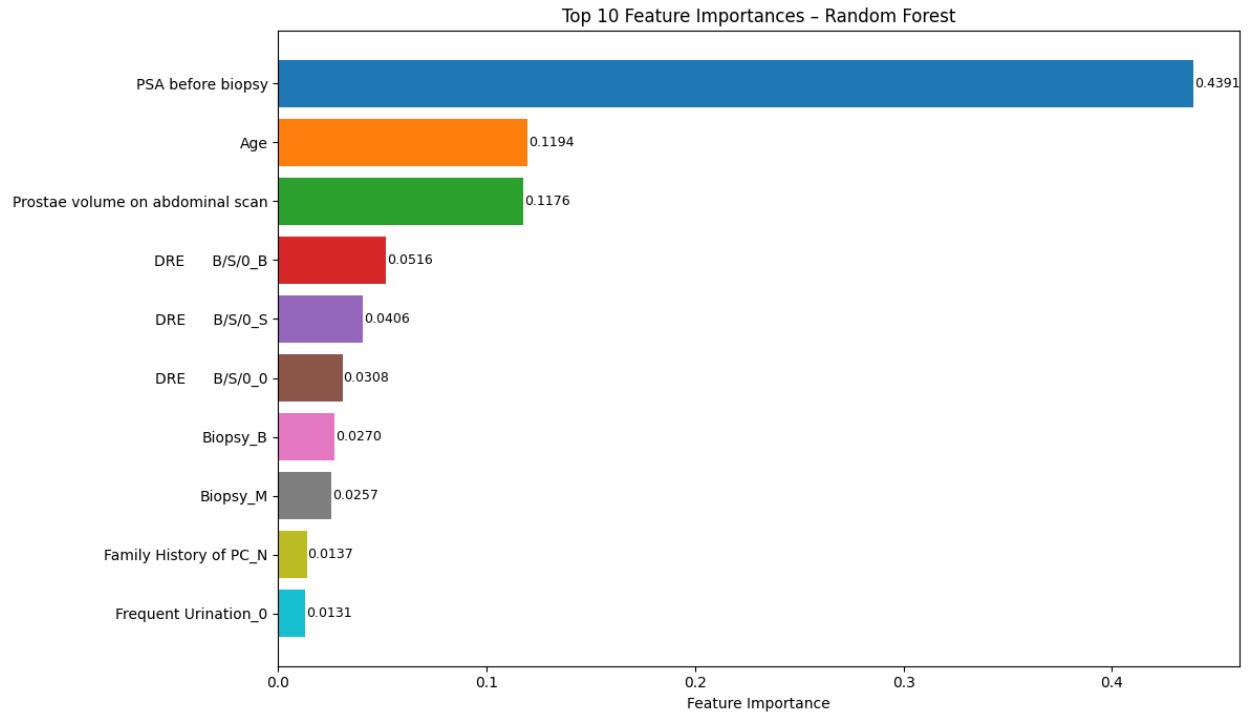


Figure 6: top 10 feature importance with Random Forest

Permutation Importance which measures the decrease in the model's score (e.g., accuracy, AUC, F1-score) when a single feature's values are randomly shuffled (or permuted) on the validation or test set was performed specifically for Voting ensemble model. It was evident that PSA before biopsy (0.0379): It is the most important feature by a large margin. Permuting this feature's values causes the largest drop in the ensemble model's performance score (0.0379 points), confirming its status as the primary predictor of prostate cancer risk.

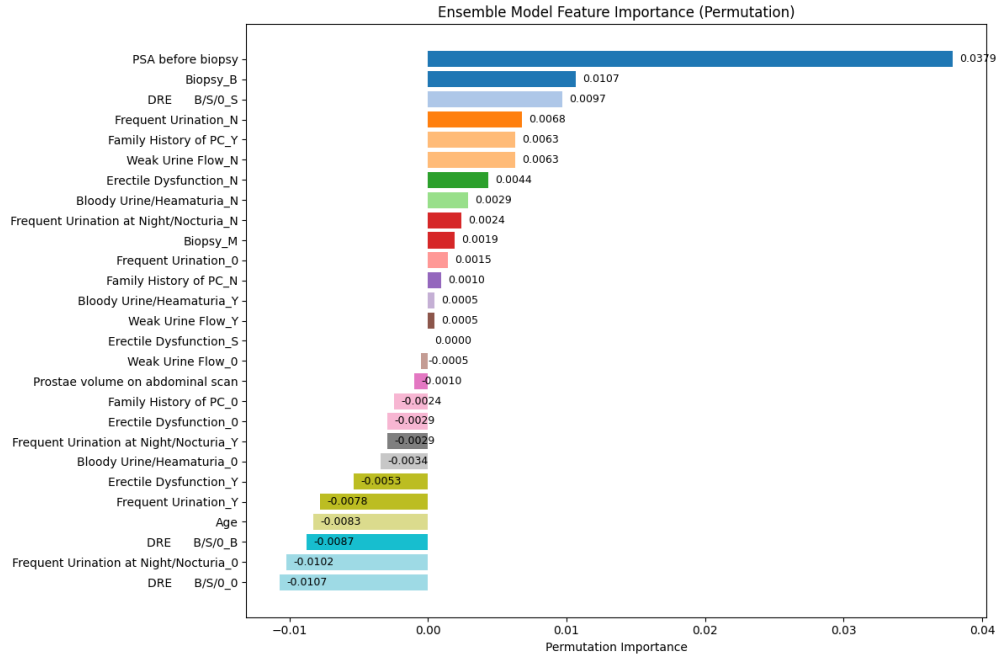


Figure 7: Feature importance permutation with Voting ensemble

CHAPTER 5

5 Constraints

Medical datasets, especially those obtained from routine clinical documentation, often contain missing or incomplete records. Such gaps can affect the accuracy and reliability of the analysis. To address this constraint, comprehensive data cleaning and preprocessing steps were incorporated into the study design. Missing values were handled appropriately through deletion of non-informative records and statistical imputation methods were applied, ensuring the dataset meets the required quality standards for valid model training and interpretation.

5.1 Conclusion

This study contributes to the growing body of evidence supporting the use of data-driven approaches in prostate cancer prediction. By leveraging clinical records and applying machine learning techniques, the study provides insights into key predictors and demonstrates the potential of computational models to support early detection and clinical decision-making.

Despite challenges such as limited data from private hospital records and missing data within the dataset, the study incorporates strong mitigation strategies including ethical clearance procedures, and robust data cleaning techniques to ensure the validity and reliability of the findings. Overall, the results underscore the value of integrating clinical expertise with predictive analytics to improve patient outcomes.

5.2 Future work

Future studies can expand on this study in several ways;

1. **Larger and Multi-site Datasets**
Incorporating data from multiple hospitals or regions would enhance model generalizability and strengthen the robustness of predictions.
2. **Incorporation of Additional Clinical Variables**
Future studies may include biochemical markers, imaging results, lifestyle factors, or genetic data to improve predictive performance.
3. **Advanced Machine Learning and Deep Learning Models**
Exploring algorithms such as gradient boosting, neural networks, or ensemble meta-learning could yield higher accuracy and richer insights.
4. **Explainable AI (XAI)**
Future work can prioritize interpretability techniques to ensure clinicians understand how predictions are generated, enhancing trust and adoption.

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